

Small Machines

A synthetic book for extraction tests

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Chapter 1

Levers and Pulleys

People have always looked for ways to move heavy things with less effort. The oldest answers are also the simplest ones: a rigid bar, a rope and a wheel. This chapter looks at two of them.

1.1 Levers

A lever is a rigid bar that turns around a fixed point called the fulcrum. When the fulcrum sits close to the load, a small push on the far end lifts a large weight. The hand moves a long way while the load moves a short way, and the computation behind it is simple enough to do in your head.

Seesaws, crowbars and bottle openers are all levers. A seesaw puts the fulcrum in the middle, so two children of similar weight can balance each other when they sit at equal

distances from the middle of the board.

A crowbar works the other way round: the fulcrum sits near the heavy end, and a long handle turns a modest push into a strong lift.¹ Builders relied on this long before anyone wrote the rule down.

Three things decide how well a lever works:

- where the fulcrum sits;
- how long each arm is; and
- how stiff the bar stays under load, which matters most when

¹ A famous boast claims that a long enough lever could move the whole Earth.

the load is heavy and the bar is thin.

1.2 Pulleys

A pulley is a wheel with a groove for a rope. One fixed pulley only changes the direction of the pull, but several pulleys working together share the load between many lengths of rope.

```
effort = load / ropes
if ropes > 1:
    print("less force, more rope")
```

Figure 1.1 — A block and tackle with four supporting ropes.

Chapter 2

Gears

A gear is a wheel with teeth. When two gears mesh, the smaller one turns faster and the larger one turns with more force, a trade that engineers call *mechanical advantage*.

Modern self-driving carts still depend on gear trains hidden inside their wheels.

Even the small self-driving carts in a warehouse count on gears to turn quick motor spins into slow, strong wheel turns.

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